

Composite functions



1

a $\frac{dy}{dx} = e^x (\sin x + \cos x)$

b $\frac{dy}{dx} = e^{\sin x} \cos x$

c $\frac{dy}{dx} = e^x (\cos 3x - 3 \sin 3x)$

d $\frac{dy}{dx} = e^{3x} \left(3 \cos \left(\frac{x}{3} \right) - \frac{1}{3} \sin \left(\frac{x}{3} \right) \right)$

e $\frac{dy}{dx} = e^{-x} (4 \cos 4x - \sin 4x)$

f $\frac{dy}{dx} = \frac{-\sin(\ln x)}{x}$

g $\frac{dy}{dx} = \frac{-\sin(\ln(-5x))}{x}$

h $\frac{dy}{dx} = -\sin x \ln x + \frac{\cos x}{x}$

i $\frac{dy}{dx} = e^x \left(\ln x + \frac{1}{x} \right)$

j $\frac{dy}{dx} = e^{2x} \left(2 \ln(4x) + \frac{1}{x} \right)$

k $\frac{dy}{dx} = \frac{1}{x \ln x}$

l $\frac{dy}{dx} = \frac{1 - 2x \ln x}{xe^{2x}}$

m $\frac{dy}{dx} = -\frac{3}{e^2 x(1 - 6x^3)}$

n $\frac{dy}{dt} = -\frac{\pi}{2} \sin \left(\frac{\pi t}{2} - \frac{\pi}{3} \right)$

o $\frac{dy}{dx} = -\cos \left(\frac{1}{x} \right) + 2x \sin \left(\frac{1}{x} \right)$

p $\frac{dy}{dx} = \frac{4}{5} \left(\cos \left(\frac{x}{5} \right) - 15e^{2x} - 10x^{-9} \right)$

q $\frac{dy}{dx} = \frac{2x - 7}{x^2 - 7x - 12} - \frac{\sqrt{11}}{2\sqrt{x}}$

r $\frac{dy}{dx} = -5 \left(e^{-5x^2} + \cos x \right)^4 \left(10xe^{-5x^2} + \sin x \right)$

s $\frac{dy}{dx} = \frac{e^{-0.2x} \left(0.2x^4 + 4x^3 - 0.2 \sin \left(\frac{\pi}{4}x \right) - \frac{\pi}{4} \cos \left(\frac{\pi}{4}x \right) \right)}{\left(\sin \left(\frac{\pi}{4}x \right) - x^4 \right)^2}$

t $\frac{dy}{dx} = e^{6x} (7 \cos x + 5 \sin x)$

u $\frac{dy}{dx} = \frac{-4 \sin \left(4x - \frac{10\pi}{11} \right) (x + 3) - 2 \cos \left(4x - \frac{10\pi}{11} \right)}{(x + 3)^3}$

2

$$\text{a } e^{5x} (5 \cos x - \sin x)$$

$$\text{c } -\sin 2x$$

$$\text{e } \frac{4 \sin x - 4x \cos x}{\sin^2(x)}$$

$$\text{g } \frac{\sin x - x \ln x \cos x}{x \sin^2(x)}$$



$$\text{b } x^3 \cos x + 3x^2 \sin x$$

$$\text{d } \frac{-2}{1 + \sin 2x}$$

$$\text{f } e^{3x} \left(3 \cos \left(5x + \frac{4\pi}{7} \right) - 5 \sin \left(5x + \frac{4\pi}{7} \right) \right)$$

3

$$\text{a } \quad \text{i } -3 \sin 3x \quad \text{ii } -\frac{3}{2}$$

$$\text{b } \quad \text{i } 3 \sin 3x \quad \text{ii } \frac{3\sqrt{3}}{2}$$

$$\text{c } \quad \text{i } 4 \cos 4x \quad \text{ii } 2\sqrt{2}$$

$$\text{d } \quad \text{i } 4 \sin 8x \quad \text{ii } 2\sqrt{2}$$

$$\text{e } \quad \text{i } -2 \sin 4x \quad \text{ii } -1$$

4

$$\text{a } \frac{dy}{dx} = -e^{\cos x} \sin x$$

$$\text{b } \frac{dy}{dx} = e^x \cos(e^x)$$

$$\text{c } \frac{dy}{dx} = e^{\cos x} (e^x \cos(e^x) - \sin x \sin(e^x))$$

5

$$\text{a } u' = 8x + e^x$$

$$\text{b } v' = -7 \sin(7x)$$

$$\text{c } \frac{d}{dx} \left(\frac{4x^2 + e^x}{\cos 7x} \right) = \frac{8x \cos 7x + e^x \cos 7x + 28x^2 \sin 7x + 7e^x \sin 7x}{\cos^2(7x)}$$

6

a No solution provided.

$$\text{b } \frac{dy}{dx} = (x + 1) e^x$$

$$\text{7 } \frac{dy}{dx} = \frac{5 (\ln(x + 5))^4}{x + 5}$$

8

$$\text{a } \frac{dy}{dx} = \frac{-2 (1 + \ln(0.3x))^{-3}}{x}$$

$$\text{b } \frac{dy}{dx} = \frac{-2e^x (1 + \ln(0.3x))^{-3}}{x} + e^x (1 + \ln(0.3x))^{-2}$$

9

a $\frac{d}{dx} (2 \ln (-3x - x^2)) = \frac{2(-3 - 2x)}{-3x - x^2}$



b $\frac{dy}{dx} = 3 \left(2 \ln (-3x - x^2) - \cos \left(\frac{\pi}{3} x \right) + \frac{1}{x^4} \right)^2 \left(\frac{2(-3 - 2x)}{-3x - x^2} + \frac{\pi}{3} \sin \left(\frac{\pi}{3} x \right) - \frac{4}{x^5} \right)$

Stationary points

10

a $f'(x) = 3e^{2x} \cos 3x + 2e^{2x} \sin 3x$

b $x = 0, \frac{\pi}{3}$

c $(0.72, 3.51)$

11

a $f'(x) = 3x^2 \ln x + x^2$

b $x = e^{-\frac{1}{3}}$

c

x	$e^{-\frac{1}{2}}$	$e^{-\frac{1}{3}}$	$e^{-\frac{1}{4}}$
$f(x)$	$-\frac{1}{2}e^{-1}$	0	$\frac{1}{4}e^{\frac{1}{2}}$

d Minimum at $x = e^{-\frac{1}{3}}$

e $-\frac{1}{3e}$

f $-\frac{1}{2e^{\frac{3}{2}}}$

12

a $x = \frac{1}{3}$

b $\frac{dy}{dx} = \frac{1 - 3x \ln 3x}{xe^{3x}}$

c No solution provided.

d

x	0.01	0.58	1.58
$\frac{dy}{dx}$	107.3	0	-0.04

e Maximum

f $(0.58, 0.10)$

13 $b = -4$ and $a = 4e$

14

a $0 < x < \frac{\pi}{2}, \pi < x < \frac{3\pi}{2}$

b $x = \frac{\pi}{4}, \frac{5\pi}{4}$

c $f\left(\frac{\pi}{4}\right) = 0$



Applications

15 $y = -\frac{\pi}{2}x + \frac{\pi^2}{4}$

16 a $y = e^{\frac{3\pi}{2}}x - \frac{3\pi e^{\frac{3\pi}{2}}}{2} + e^{\frac{3\pi}{2}} + 3$

b $y = x + 1 - \frac{3\pi}{2}$

17 a $P(0) = 4420$ b $P'(0) = 291$ c $P'\left(\frac{7}{12}\right) = 318$ d $P'(10) = 1036$

18 a $x(0) = 29 \text{ m}$ b $v(t) = (2t + \cos(t - 4\pi) - 1) \text{ m/s}$
c $a(t) = (2 - \sin(t - 4\pi)) \text{ m/s}^2$

19 a $x = 0$ b $\frac{dx}{dt} = 2\pi e^{2t} \cos(\pi e^{2t})$
c $t = \frac{1}{2} \ln \frac{3}{2}, \frac{1}{2} \ln \frac{5}{2}$ d $x = -1$ means 1 unit left of the origin.
e $x = 1$ means 1 unit right of the origin.